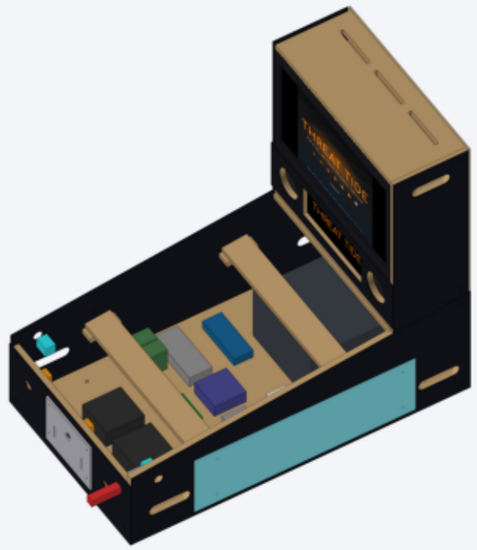
Threat Tide pinball cabinet

32" playfield, 24" backglass + 12.3" DMD bar, 18 mm plywood, knock-down for van / ute transport

Assembled



Inside (left side hidden)



Transport modules




```
Overall: 600 W x 1155 D (incl. plunger) x 1572 H mm
Lockbar height 911 mm, playfield slope 6.5 deg (built into the sides)

Transport modules (empty weight estimate, plywood 600 kg/m3):
  body       630 x  1075 x   381 mm   ~30.6 kg wood
  backbox    606 x   245 x   586 mm   ~11.3 kg wood
  legs       600 x   920 x   608 mm   ~13.6 kg wood
  + components: body ~13 kg (monitor, PC, contactors, shakers), backbox ~6 kg

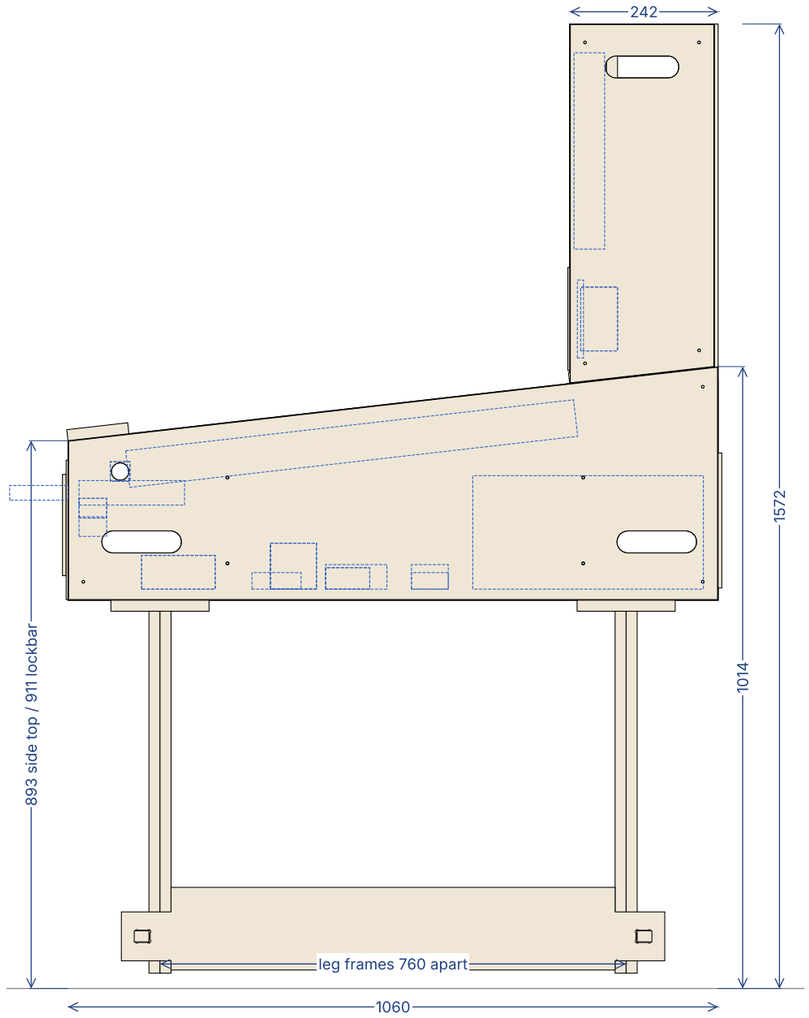
Clearance check: passed - no part or component collisions

Sheets: 2 x 18 mm plywood, 1 x 6 mm plywood, 1 x 6 mm MDF, 1 x 3 mm clear cast acrylic, 1 x 3 mm ACM (aluminium composite) + printed vinyl
DXF layers: CUT_OUTSIDE, CUT_INSIDE, DRILL, POCKET_6 (6 deep), POCKET_6_BACK (other face),
            ENGRAVE, MARK (layout lines, do not cut), NOTES
```

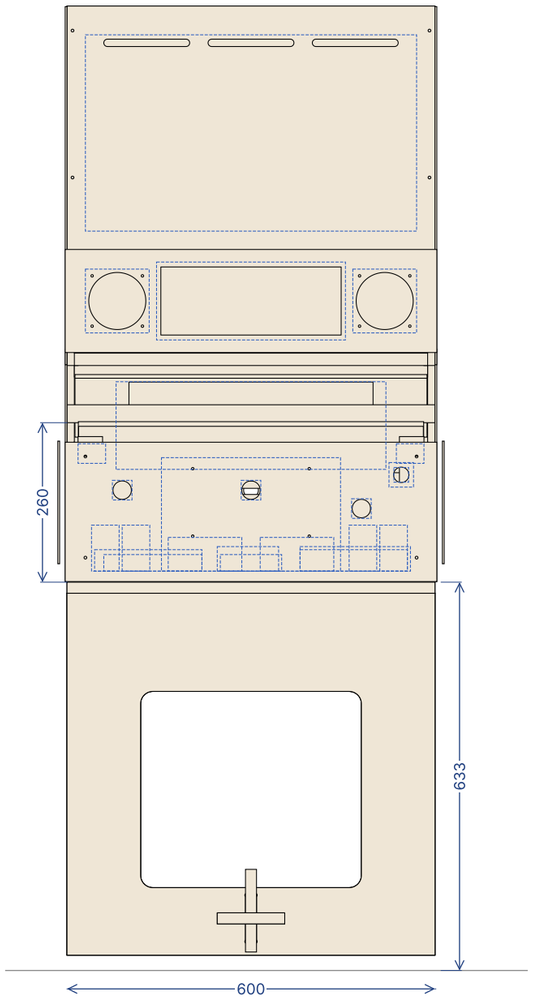
General arrangement

Side, front and plan views at 1:10 on A3. Ground = 0.

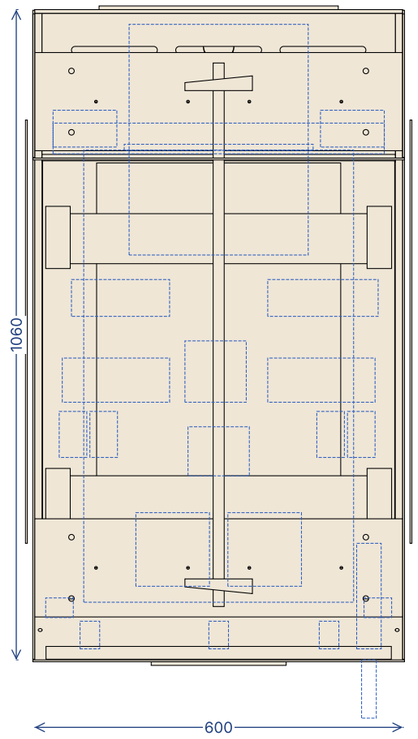
Side elevation (from left) 1:10



Front elevation 1:10



Plan 1:10

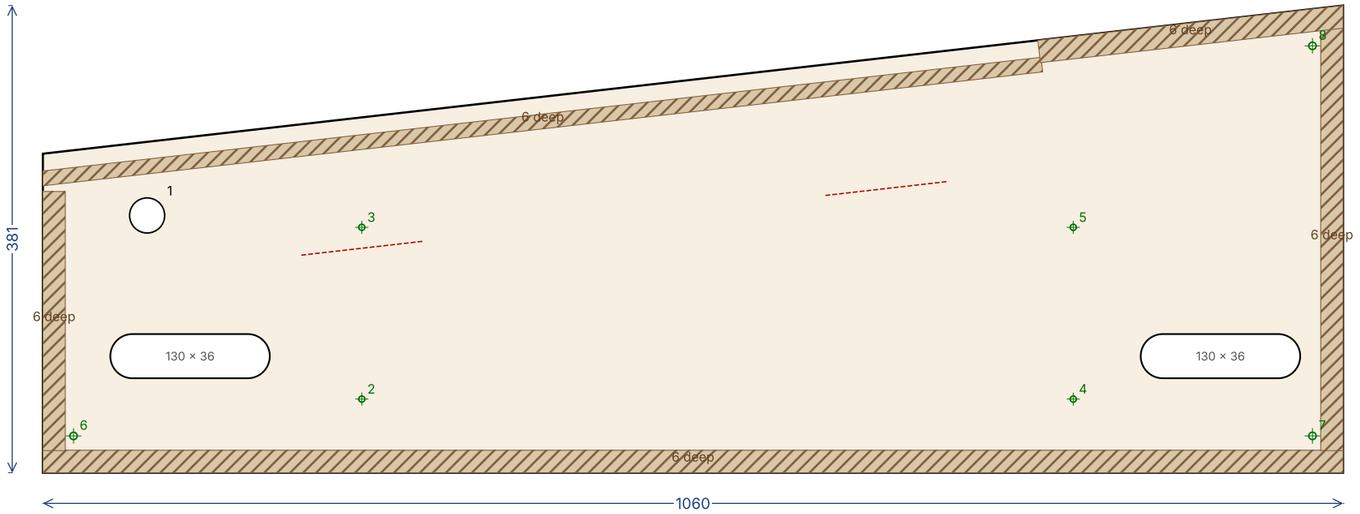


Blue dashed = bought components (monitor, PC, buttons, plunger, contactors, shakers, supplies, speakers). Clearances are checked automatically in code.

P01 Body side L

18 mm plywood | qty 1 | module: body

scale 1:5 at A3 (origin = lower-left corner)



Pockets are on the inside face. R is the mirror of L.

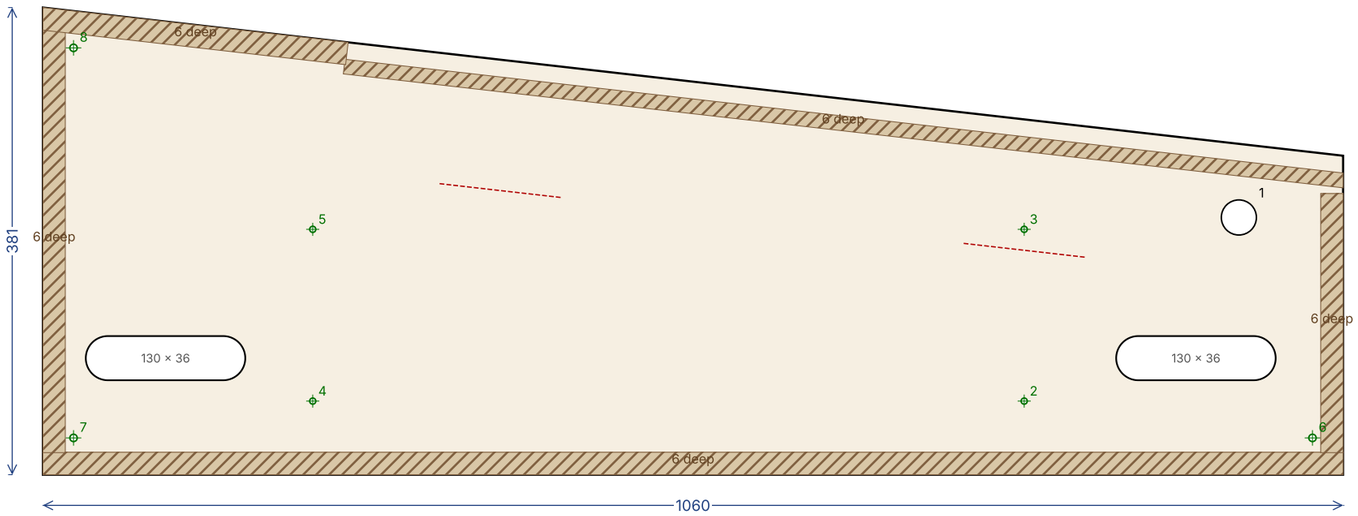
#	dia	X	Y	op
1	28.6	85.0	209.7	profile
2	5.0	260.0	60.0	drill
3	5.0	260.0	200.0	drill
4	5.0	840.0	60.0	drill
5	5.0	840.0	200.0	drill
6	6.0	25.0	30.0	drill
7	6.0	1035.0	30.0	drill
8	6.0	1035.0	347.9	drill

Pockets (bounding boxes):
- 0.0,0.0 to 1060.0,18.3 6 deep top
- 0.0,18.0 to 18.3,229.2 6 deep top
- 1041.7,18.0 to 1060.0,362.6 6 deep top
- 0.0,233.8 to 815.0,338.6 6 deep top
- 811.1,334.2 to 1060.0,380.8 6 deep top

P02 Body side R

18 mm plywood | qty 1 | module: body | drawn mirrored: shows the cut face (same as the DXF)

scale 1:5 at A3 (origin = lower-left corner)



Pockets are on the inside face. R is the mirror of L.

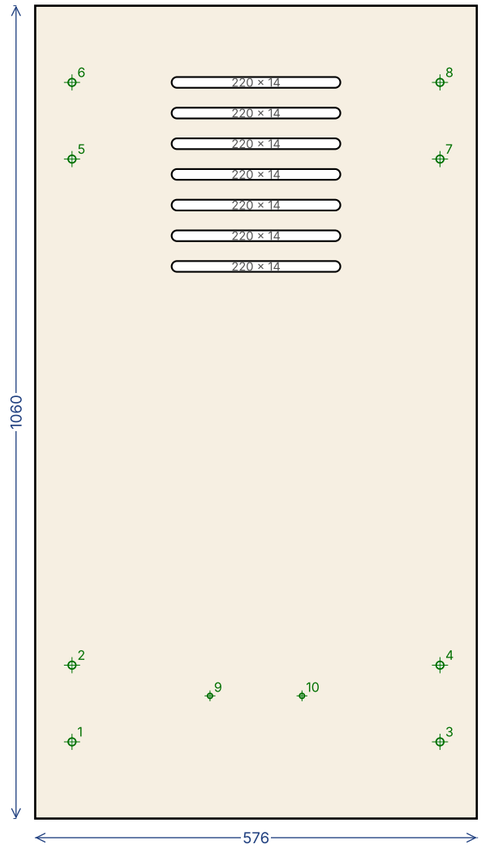
#	dia	X	Y	op
1	28.6	975.0	209.7	profile
2	5.0	800.0	60.0	drill
3	5.0	800.0	200.0	drill
4	5.0	220.0	60.0	drill
5	5.0	220.0	200.0	drill
6	6.0	1035.0	30.0	drill
7	6.0	25.0	30.0	drill
8	6.0	25.0	347.9	drill

Pockets (bounding boxes):
- 0.0,0.0 to 1060.0,18.3 6 deep top
- 1041.7,18.0 to 1060.0,229.2 6 deep top
- 0.0,18.0 to 18.3,362.6 6 deep top
- 245.0,233.8 to 1060.0,338.6 6 deep top
- 0.0,334.2 to 248.9,380.8 6 deep top

P03 Body floor

18 mm plywood | qty 1 | module: body

scale 1:8 at A3 (origin = lower-left corner)



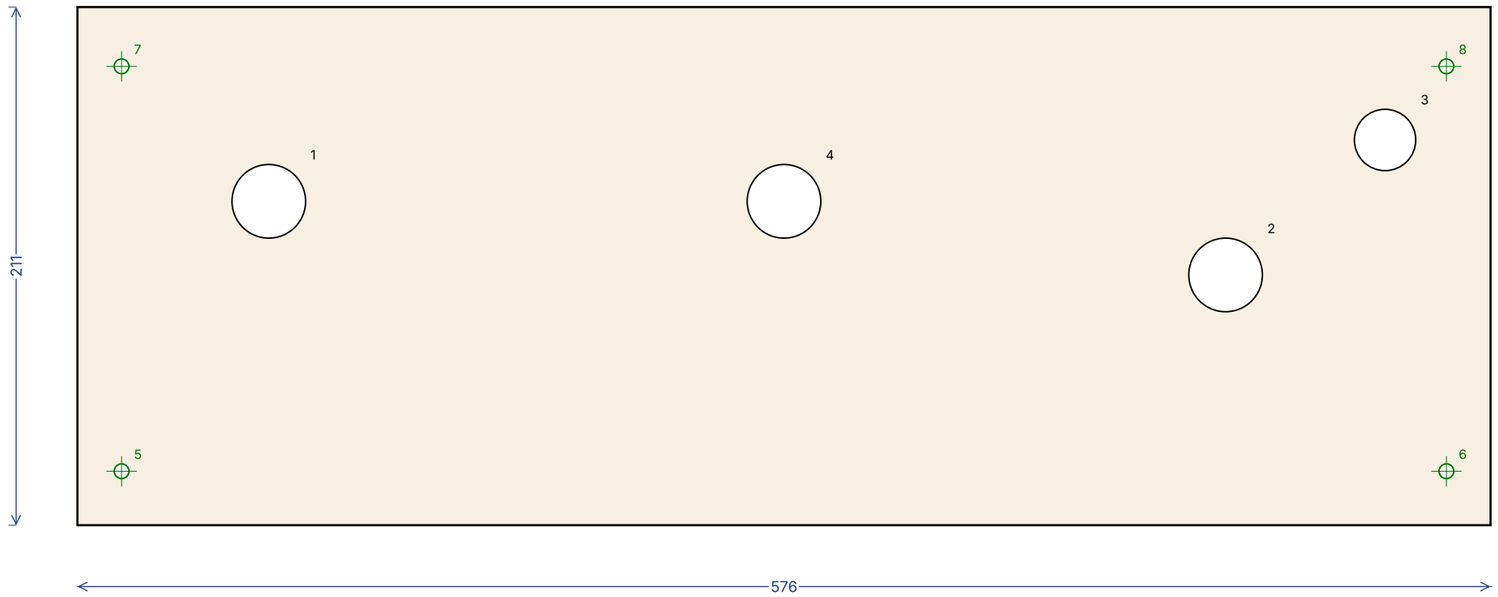
Tee nuts (M8) pressed in from the top for the leg bolts.

#	dia	X	Y	op
1	10.0	48.0	100.0	drill
2	10.0	48.0	200.0	drill
3	10.0	528.0	100.0	drill
4	10.0	528.0	200.0	drill
5	10.0	48.0	860.0	drill
6	10.0	48.0	960.0	drill
7	10.0	528.0	860.0	drill
8	10.0	528.0	960.0	drill
9	6.5	228.0	160.0	drill
10	6.5	348.0	160.0	drill

P04 Body front

18 mm plywood | qty 1 | module: body

scale 1:2.5 at A3 (origin = lower-left corner)



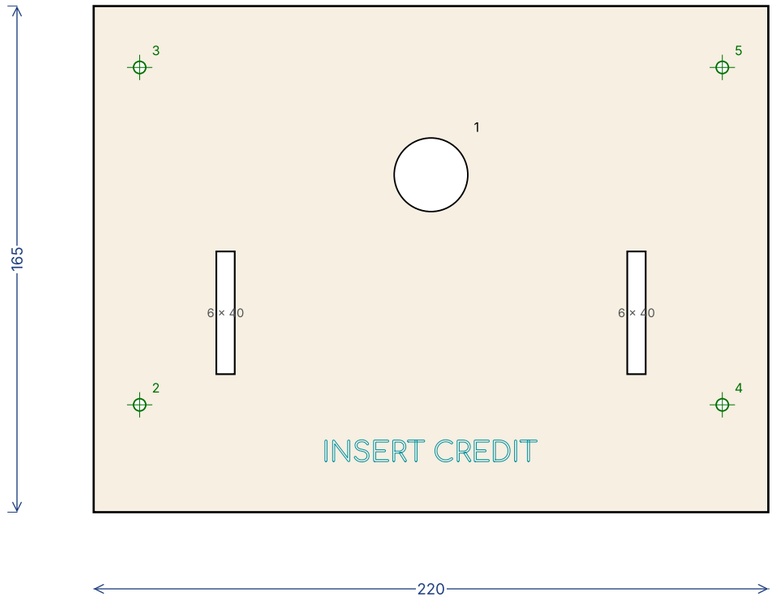
Plunger flange holes: drill to your plunger kit on assembly.

#	dia	X	Y	op
1	30.0	78.0	132.0	profile
2	30.0	468.0	102.0	profile
3	25.0	533.0	157.0	profile
4	30.0	288.0	132.0	profile
5	6.0	18.0	22.0	drill
6	6.0	558.0	22.0	drill
7	6.0	18.0	187.0	drill
8	6.0	558.0	187.0	drill

P05 Coin door overlay

6 mm plywood | qty 1 | module: body

scale 1:2 at A3 (origin = lower-left corner)



Paint silver or wrap in brushed vinyl. Slots are decorative.

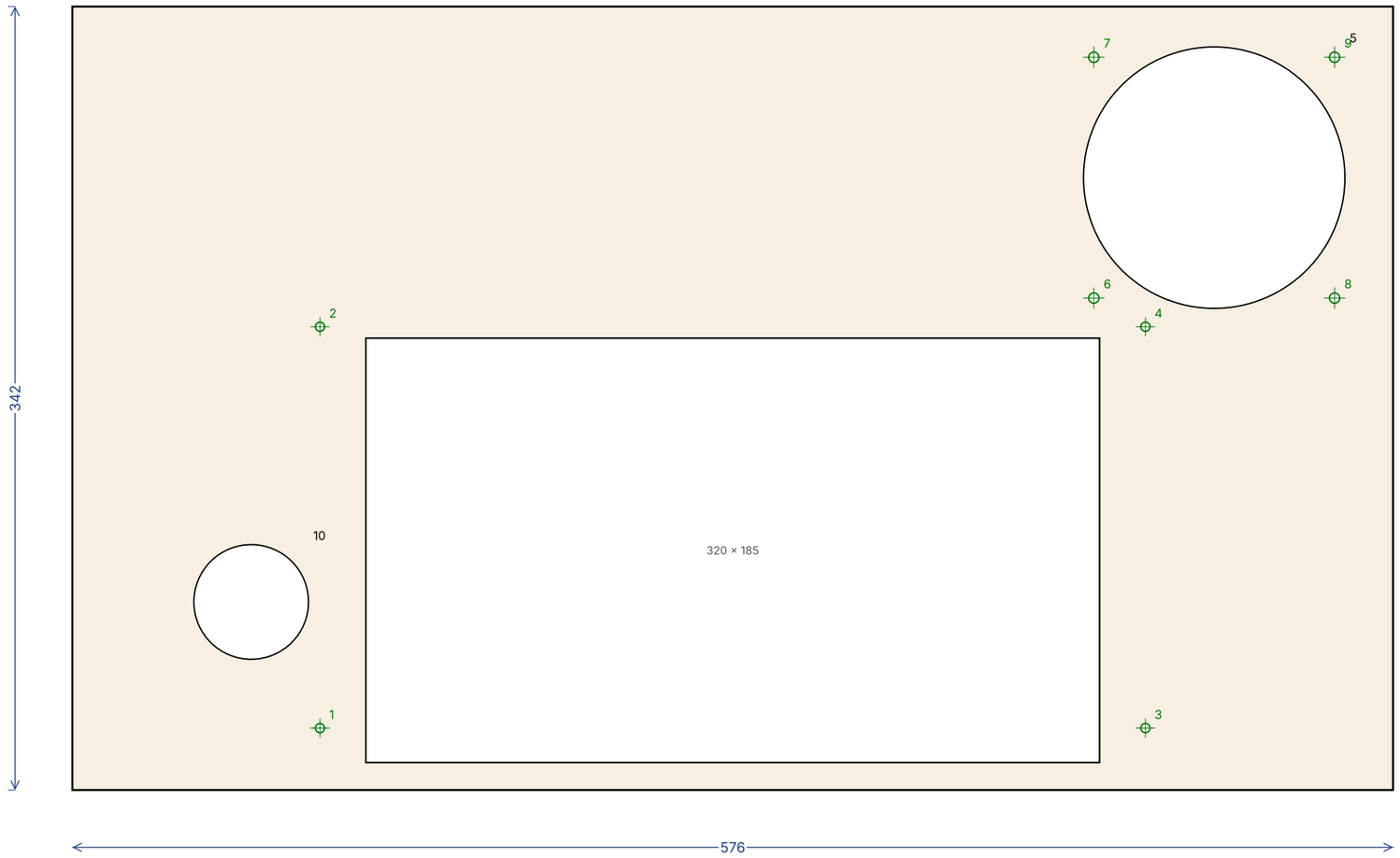
#	dia	X	Y	op
1	24.0	110.0	110.0	profile
2	4.0	15.0	35.0	drill
3	4.0	15.0	145.0	drill
4	4.0	205.0	35.0	drill
5	4.0	205.0	145.0	drill

P06 Body back

18 mm plywood | qty 1 | module: body

scale 1:2.5 at A3 (origin = lower-left corner)

#	dia	X	Y	op
1	4.0	108.0	27.0	drill
2	4.0	108.0	202.0	drill
3	4.0	468.0	27.0	drill
4	4.0	468.0	202.0	drill
5	114.0	498.0	267.0	profile
6	4.5	445.5	214.5	drill
7	4.5	445.5	319.5	drill
8	4.5	550.5	214.5	drill
9	4.5	550.5	319.5	drill
10	50.0	78.0	82.0	profile



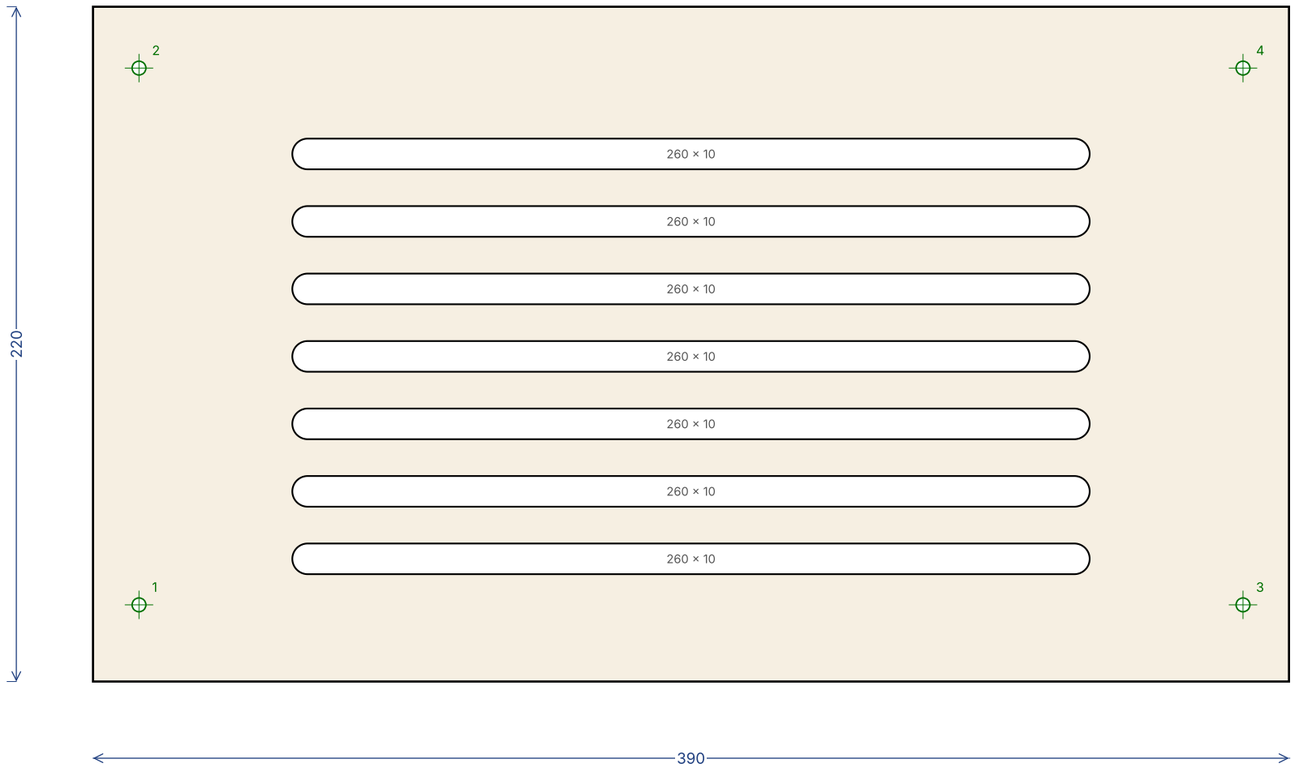
P07 Back service cover

6 mm plywood | qty 1 | module: body

scale 1:2 at A3 (origin = lower-left corner)

Screws on with 4 x M4 into the back panel. Vents for the PC.

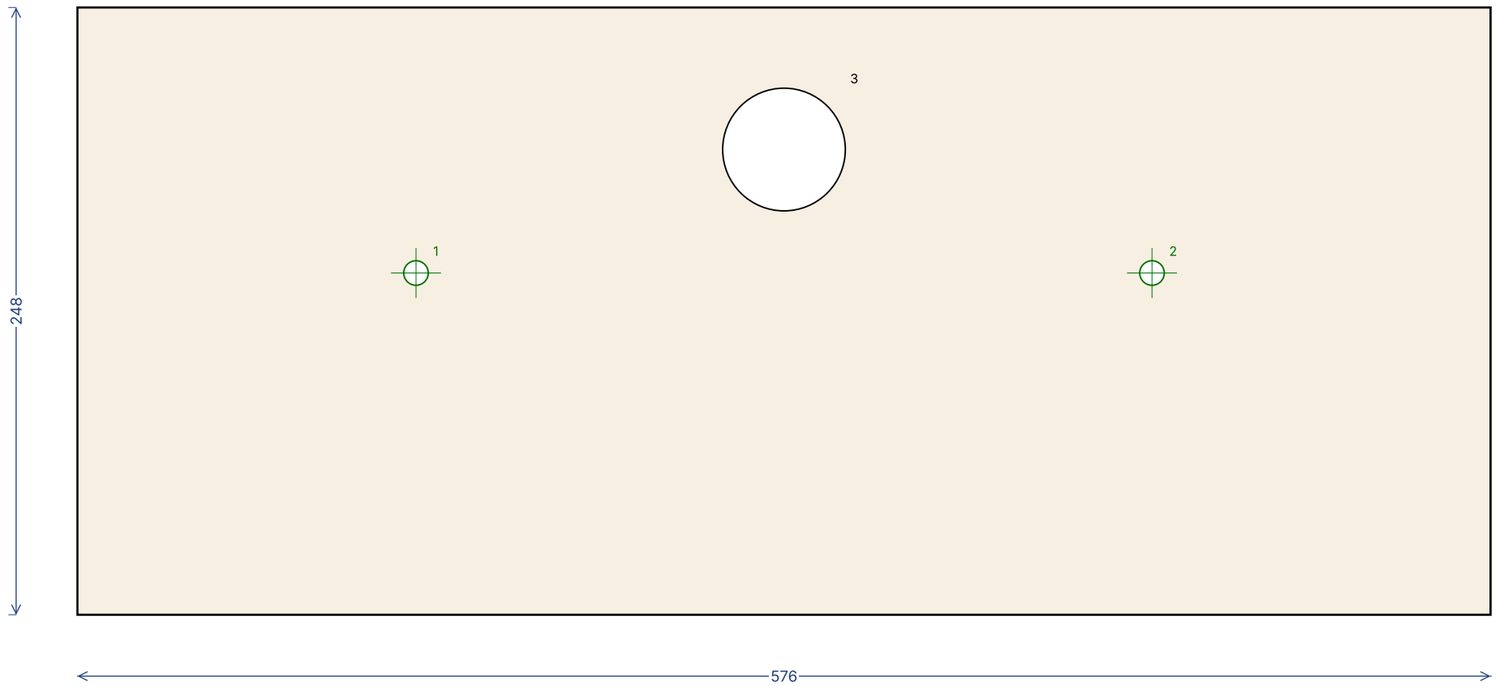
#	dia	X	Y	op
1	4.5	15.0	25.0	drill
2	4.5	15.0	200.0	drill
3	4.5	375.0	25.0	drill
4	4.5	375.0	200.0	drill



P08 Rear deck

18 mm plywood | qty 1 | module: body

scale 1:2.5 at A3 (origin = lower-left corner)



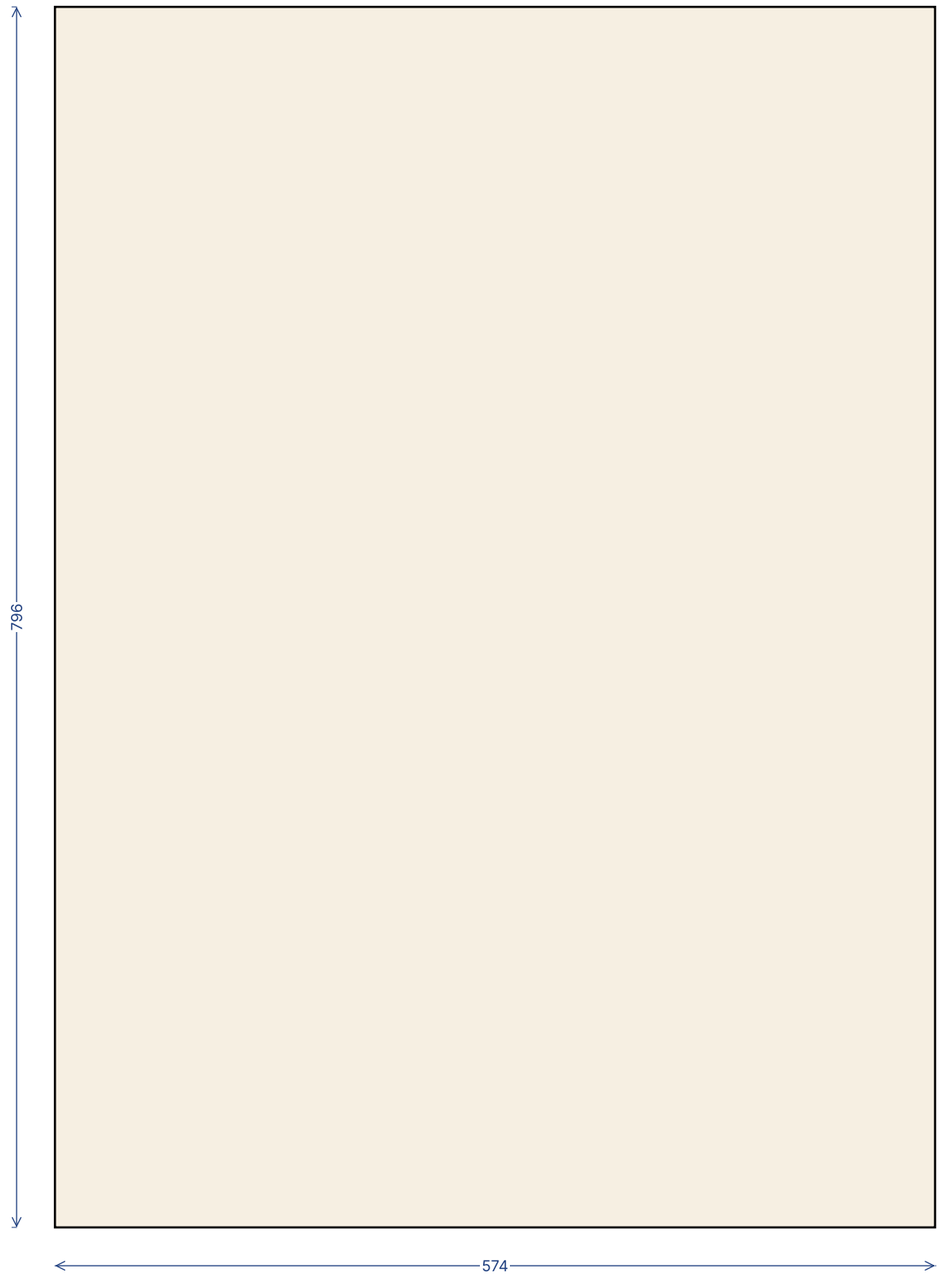
Tee nuts on the underside for the backbox bolts.

#	dia	X	Y	op
1	10.0	138.0	139.3	drill
2	10.0	438.0	139.3	drill
3	50.0	288.0	189.6	profile

P09 Playfield glass

5 mm toughened glass | qty 1 | module: body | supplier cut

scale 1:4 at A3 (origin = lower-left corner)



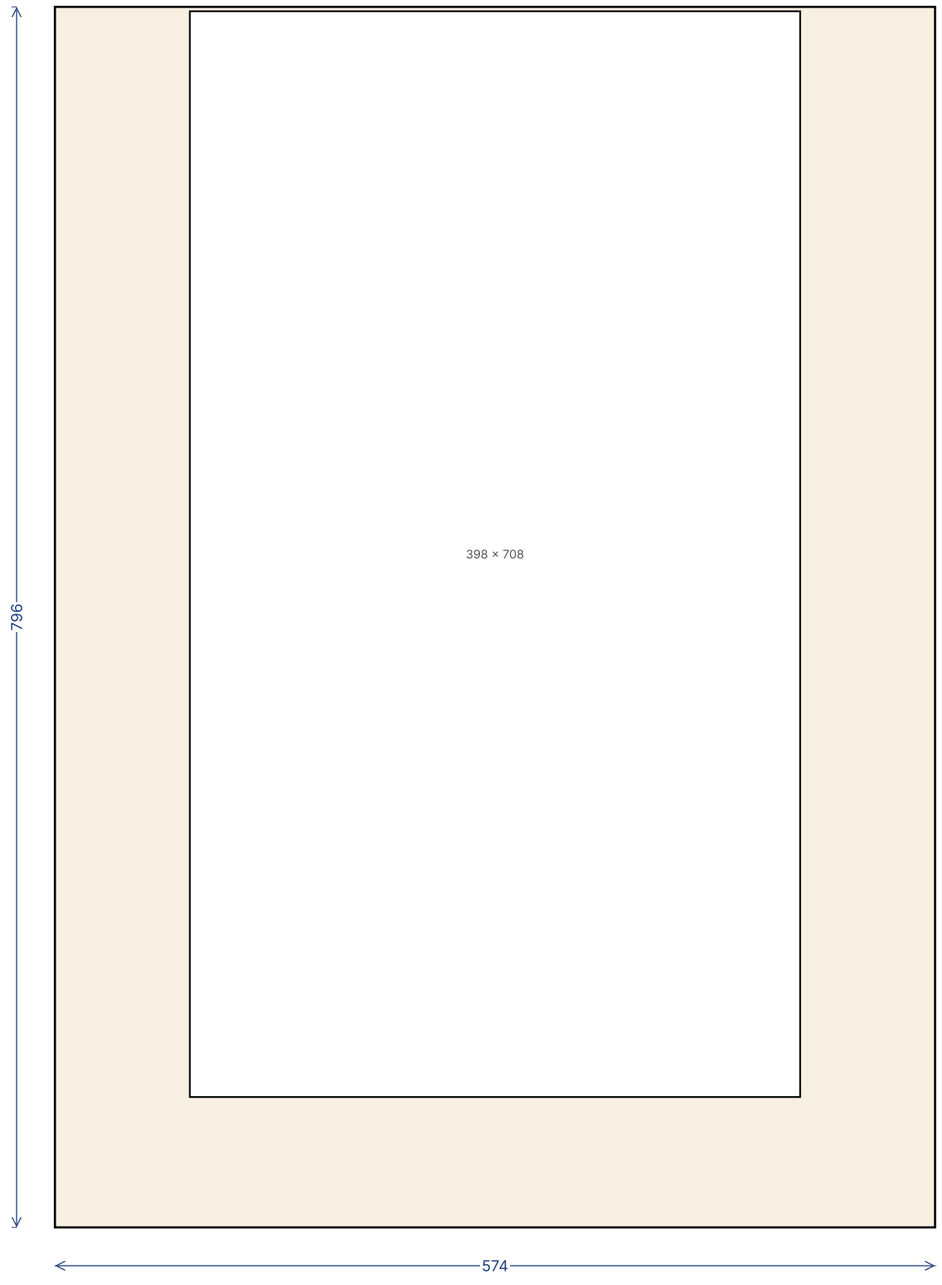

```
5 mm toughened, edges arrised. Order cut to size
from a glazier.
```

P10 Playfield mask

6 mm MDF | qty 1 | module: body

scale 1:4 at A3 (origin = lower-left corner)

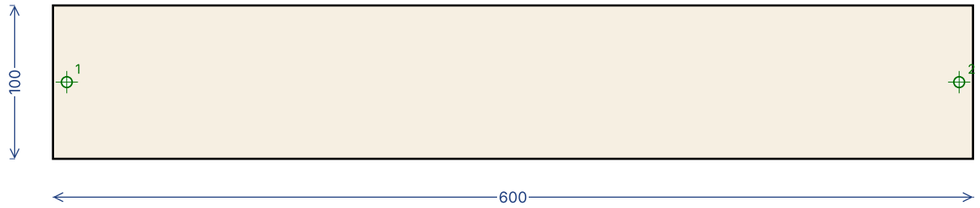
Paint matt black. Window = TV visible area; check against your TV and adjust TV_VIEW.



P11 Lockbar

18 mm plywood | qty 1 | module: body

scale 1:4 at A3 (origin = lower-left corner)



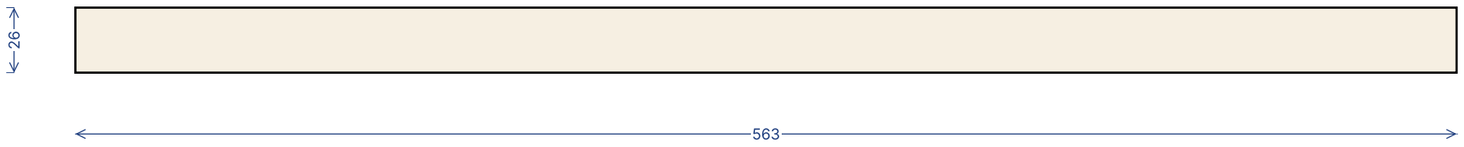
Round over the front edge (R9) and wrap in brushed vinyl or cap with aluminium angle.

#	dia	X	Y	op
1	7.0	9.0	50.0	drill
2	7.0	591.0	50.0	drill

P12 Lockbar stop strip

18 mm plywood | qty 1 | module: body

scale 1:2.5 at A3 (origin = lower-left corner)



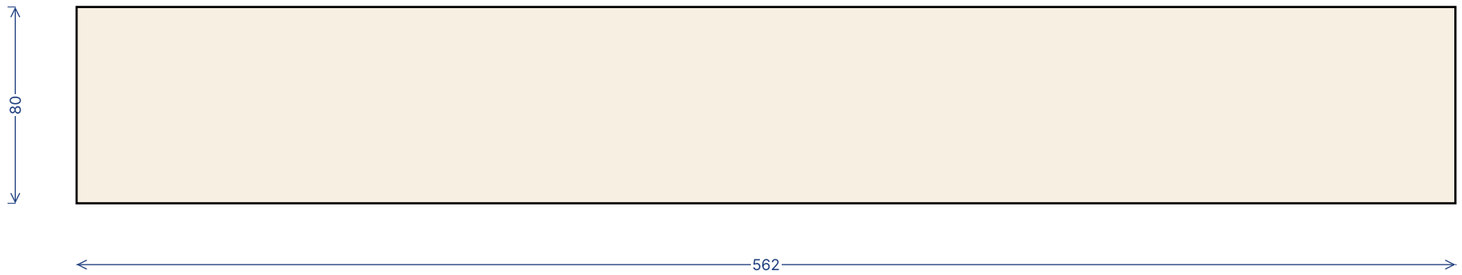
Glued and screwed under the lockbar; holds the glass and mask in.

P13 TV crossbar 1

18 mm plywood | qty 1 | module: body

scale 1:2.5 at A3 (origin = lower-left corner)

Lift-out. 3 mm foam tape on top presses the TV up against the mask.

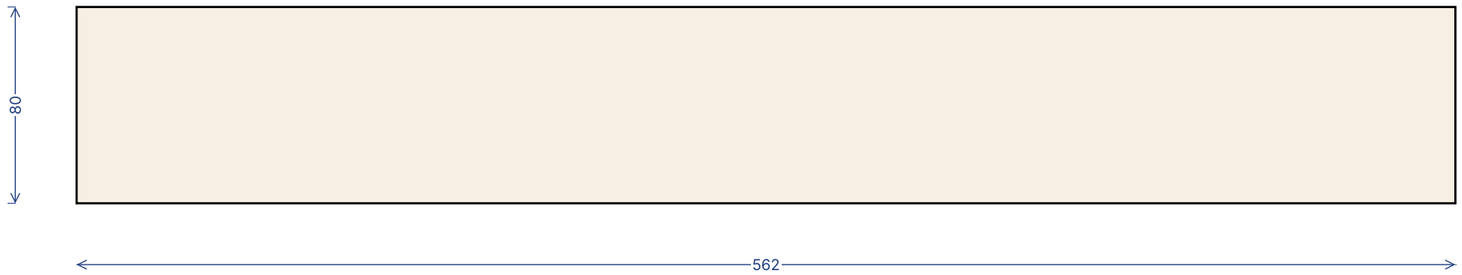


P14 TV crossbar 2

18 mm plywood | qty 1 | module: body

scale 1:2.5 at A3 (origin = lower-left corner)

Lift-out. 3 mm foam tape on top presses the TV up against the mask.

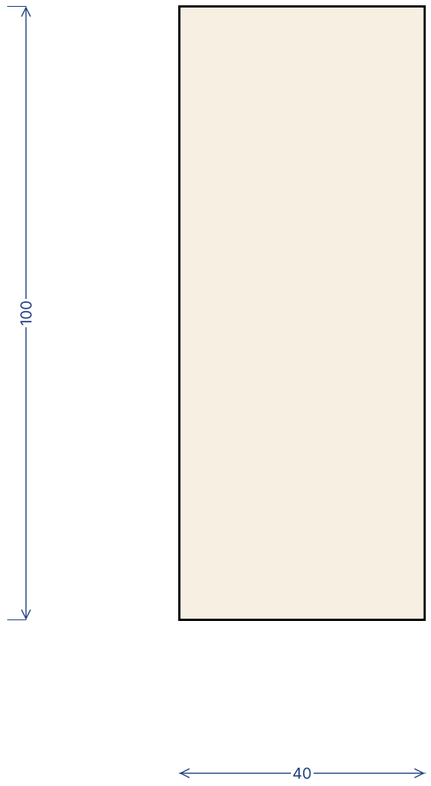


P15 Crossbar cleat

18 mm plywood | qty 4 | module: body

scale 1:1 at A3 (origin = lower-left corner)

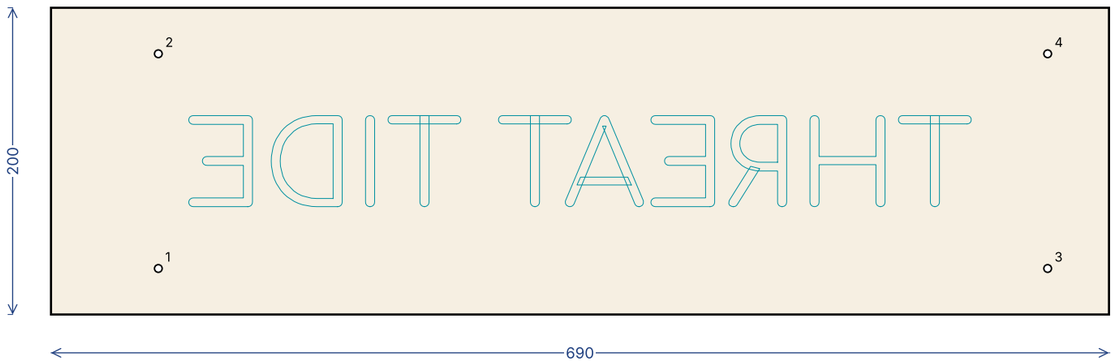
Glue and screw to the sides on the layout marks.



P16 Side light panel L

3 mm clear cast acrylic | qty 1 | module: body

scale 1:4 at A3 (origin = lower-left corner)



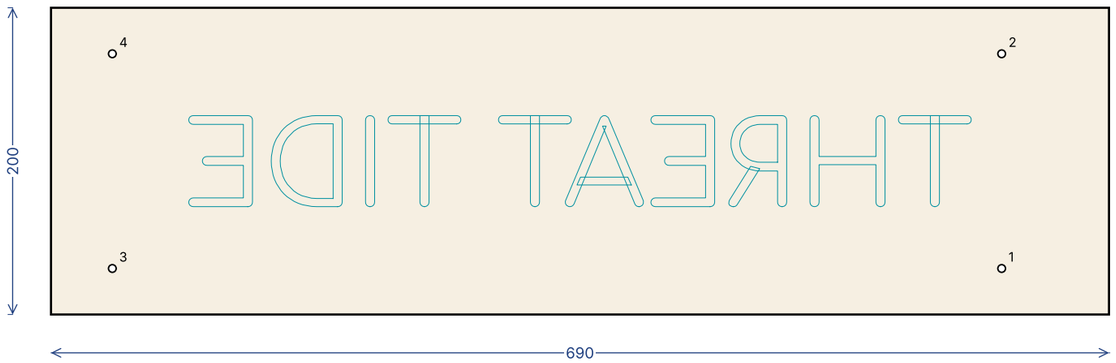
3 mm clear cast acrylic, engraved on the face against the cabinet (text is pre-mirrored). 12 mm standoffs; LED strip in a channel under the bottom edge.

#	dia	X	Y	op
1	5.0	70.0	30.0	profile
2	5.0	70.0	170.0	profile
3	5.0	650.0	30.0	profile
4	5.0	650.0	170.0	profile

P17 Side light panel R

3 mm clear cast acrylic | qty 1 | module: body | drawn mirrored: shows the cut face (same as the DXF)

scale 1:4 at A3 (origin = lower-left corner)



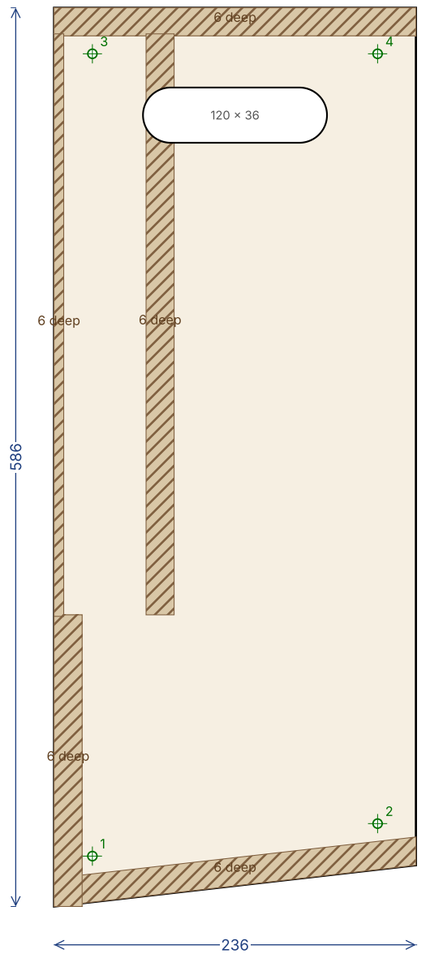
3 mm clear cast acrylic, engraved on the face against the cabinet (text is pre-mirrored). 12 mm standoffs; LED strip in a channel under the bottom edge.

#	dia	X	Y	op
1	5.0	620.0	30.0	profile
2	5.0	620.0	170.0	profile
3	5.0	40.0	30.0	profile
4	5.0	40.0	170.0	profile

P18 Backbox side L

18 mm plywood | qty 1 | module: backbox

scale 1:4 at A3 (origin = lower-left corner)



Pockets on the inside face. R is the mirror of L.

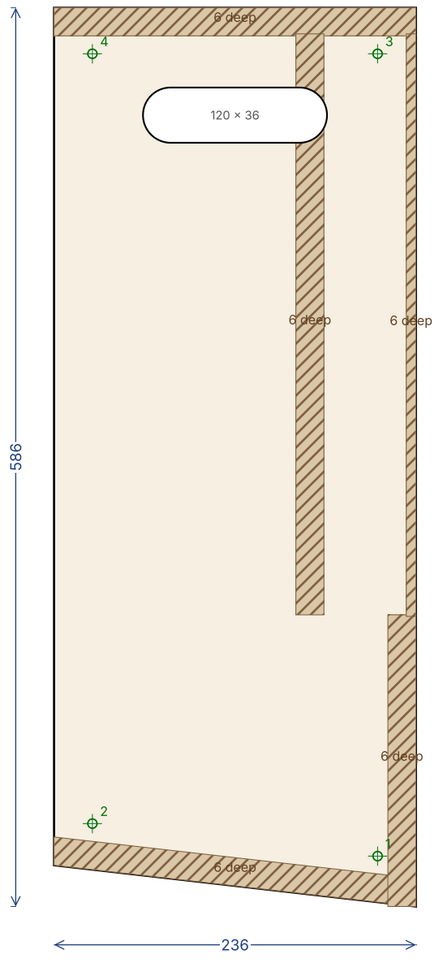
#	dia	X	Y	op
1	6.0	25.0	32.8	drill
2	6.0	211.0	54.0	drill
3	6.0	25.0	556.0	drill
4	6.0	211.0	556.0	drill

Pockets (bounding boxes):
- 0.0,0.0 to 236.0,45.3 6 deep top
- 0.0,567.7 to 236.0,586.0 6 deep top
- 0.0,0.0 to 18.3,190.2 6 deep top
- 0.0,189.2 to 6.3,569.0 6 deep top
- 60.0,190.2 to 78.3,569.0 6 deep top

P19 Backbox side R

18 mm plywood | qty 1 | module: backbox | drawn mirrored: shows the cut face (same as the DXF)

scale 1:4 at A3 (origin = lower-left corner)



Pockets on the inside face. R is the mirror of L.

#	dia	X	Y	op
1	6.0	211.0	32.8	drill
2	6.0	25.0	54.0	drill
3	6.0	211.0	556.0	drill
4	6.0	25.0	556.0	drill

Pockets (bounding boxes):

- 0.0,0.0 to 236.0,45.3 6 deep top
- 0.0,567.7 to 236.0,586.0 6 deep top
- 217.7,0.0 to 236.0,190.2 6 deep top
- 229.7,189.2 to 236.0,569.0 6 deep top
- 157.7,190.2 to 176.0,569.0 6 deep top

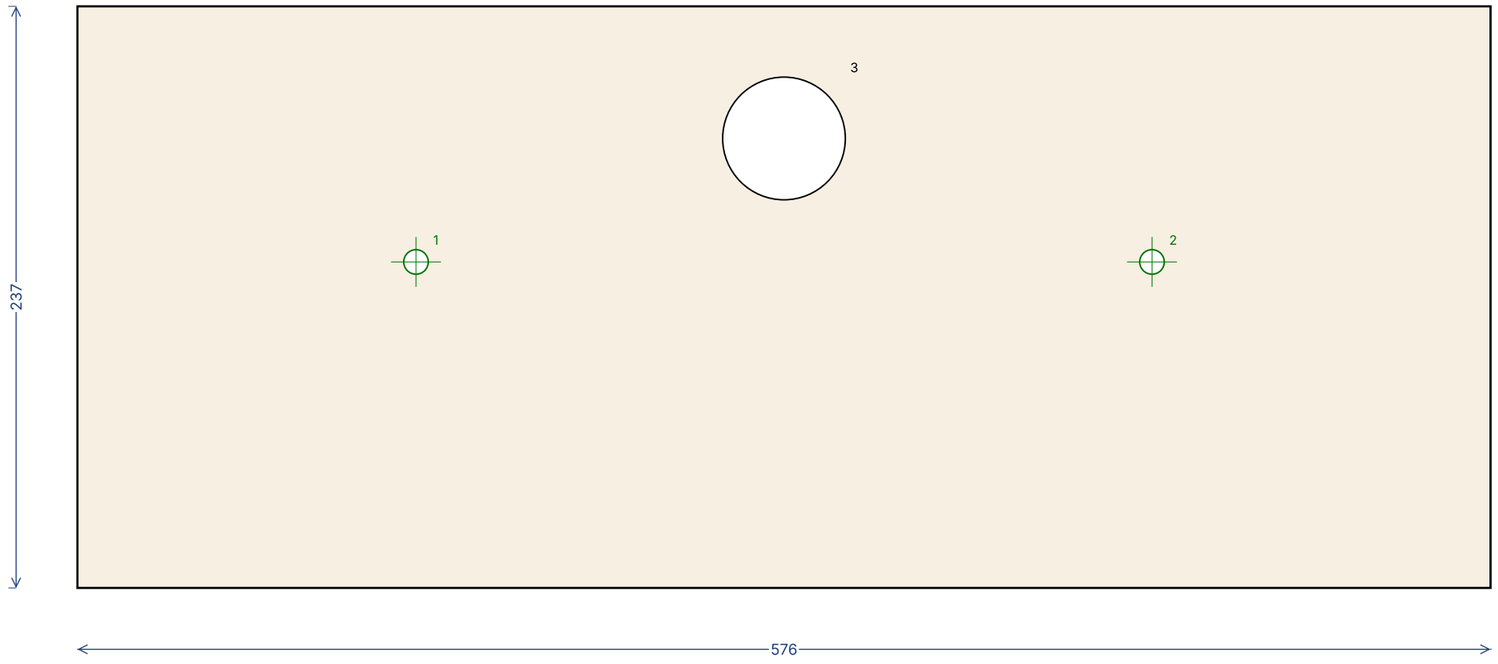
P20 Backbox bottom

18 mm plywood | qty 1 | module: backbox

scale 1:2.5 at A3 (origin = lower-left corner)

M8 bolts down through here into the deck tee nuts.

#	dia	X	Y	op
1	10.0	138.0	132.9	drill
2	10.0	438.0	132.9	drill
3	50.0	288.0	183.2	profile



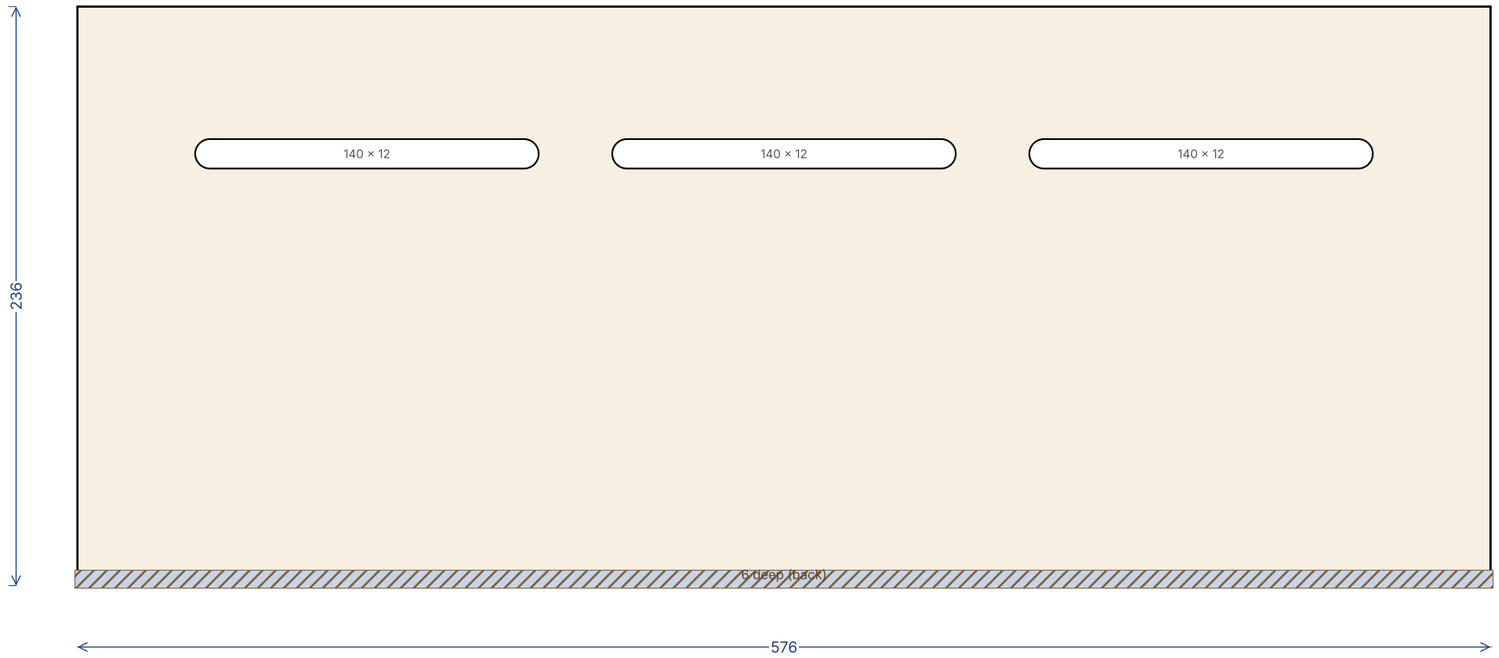
P21 Backbox top

18 mm plywood | qty 1 | module: backbox

scale 1:2.5 at A3 (origin = lower-left corner)

Bezel groove is on the underside (flip the part for that op).

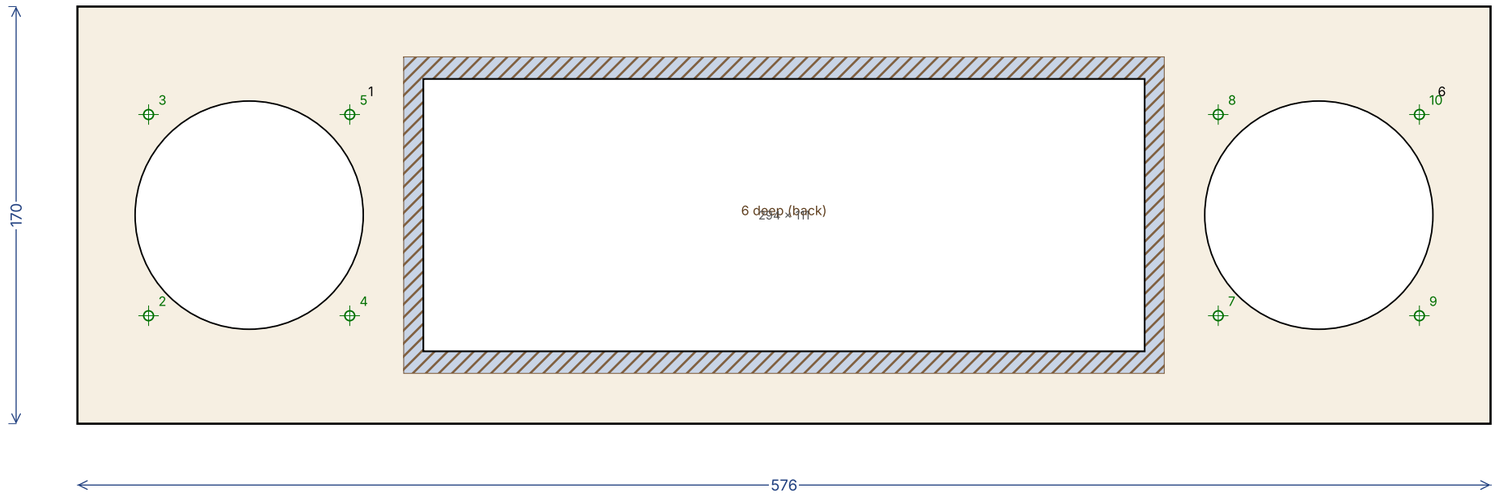
Pockets (bounding boxes):
  -1.0,-1.0 to 577.0,6.3 6 deep bottom



P22 Speaker / DMD panel

18 mm plywood | qty 1 | module: backbox

scale 1:2.5 at A3 (origin = lower-left corner)



DMD pocket is on the BACK face (flip the part for that op): the LCD sits in it.

#	dia	X	Y	op
1	93.0	70.0	85.0	profile
2	4.0	29.0	44.0	drill
3	4.0	29.0	126.0	drill
4	4.0	111.0	44.0	drill
5	4.0	111.0	126.0	drill
6	93.0	506.0	85.0	profile
7	4.0	465.0	44.0	drill
8	4.0	465.0	126.0	drill
9	4.0	547.0	44.0	drill
10	4.0	547.0	126.0	drill

Pockets (bounding boxes):
  133.0,20.5 to 443.0,149.5 6 deep bottom

P23 Monitor bezel

6 mm MDF | qty 1 | module: backbox

scale 1:2.5 at A3 (origin = lower-left corner)

Paint matt black. Window = 24" visible area; check against your monitor.



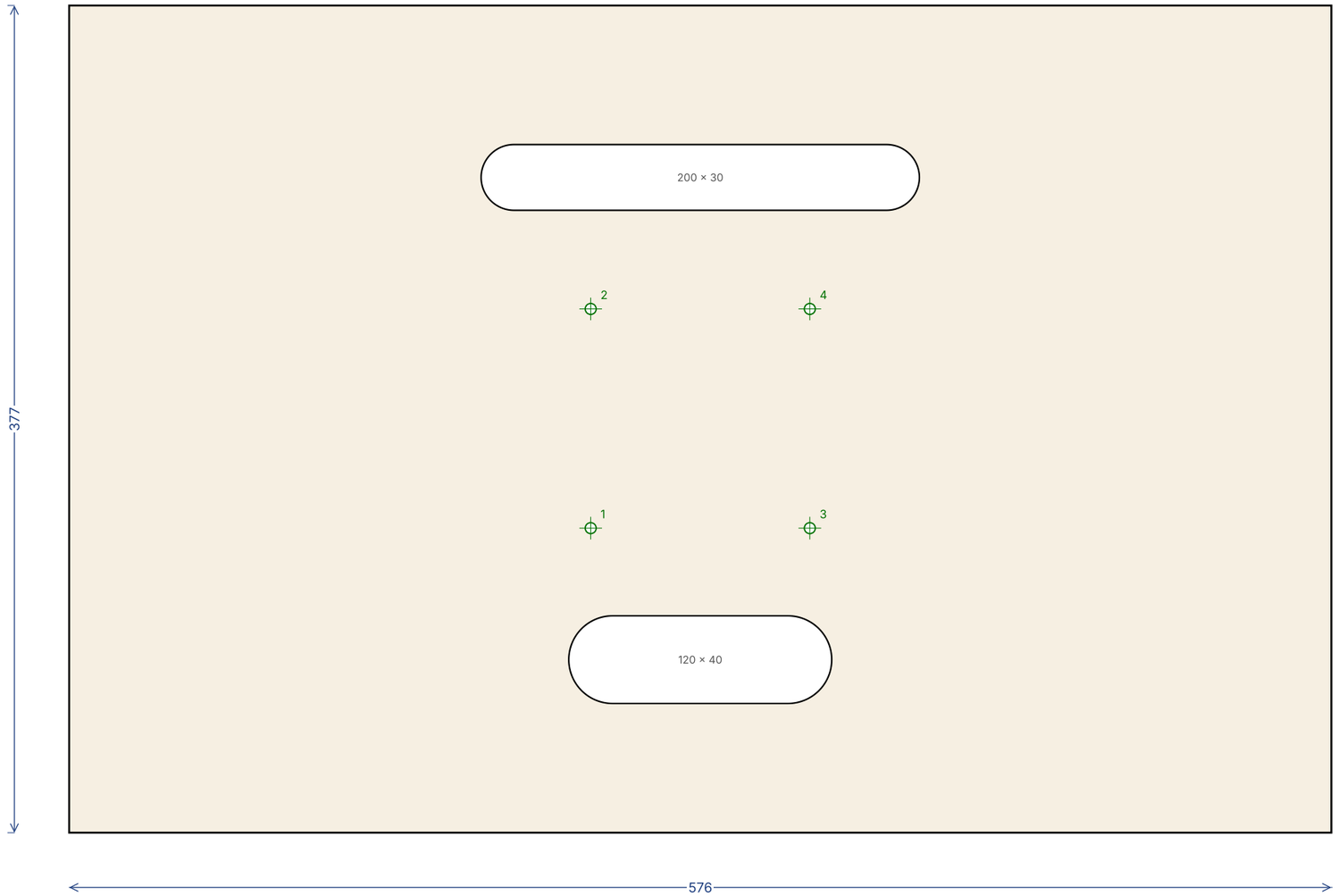
P24 Monitor board

18 mm plywood | qty 1 | module: backbox

scale 1:2.5 at A3 (origin = lower-left corner)

VESA 100 holes; slots for cables and airflow.

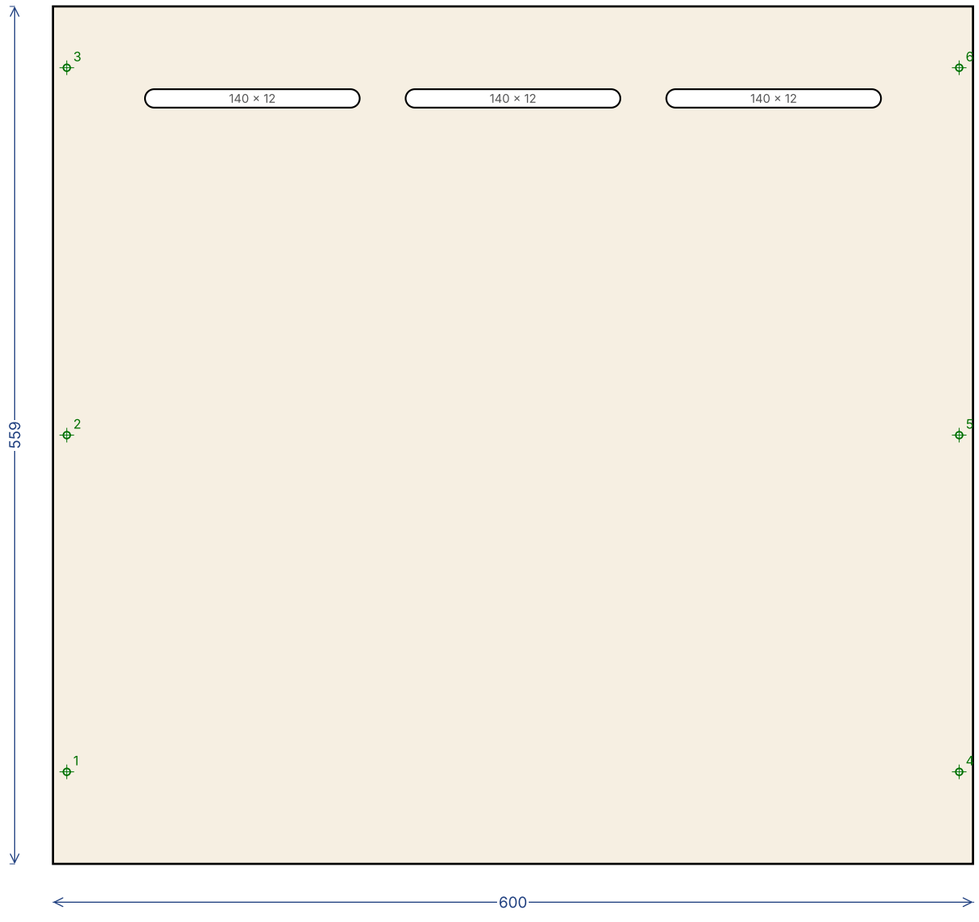
#	dia	X	Y	op
1	5.0	238.0	138.9	drill
2	5.0	238.0	238.9	drill
3	5.0	338.0	138.9	drill
4	5.0	338.0	238.9	drill



P25 Backbox back

6 mm plywood | qty 1 | module: backbox

scale 1:4 at A3 (origin = lower-left corner)



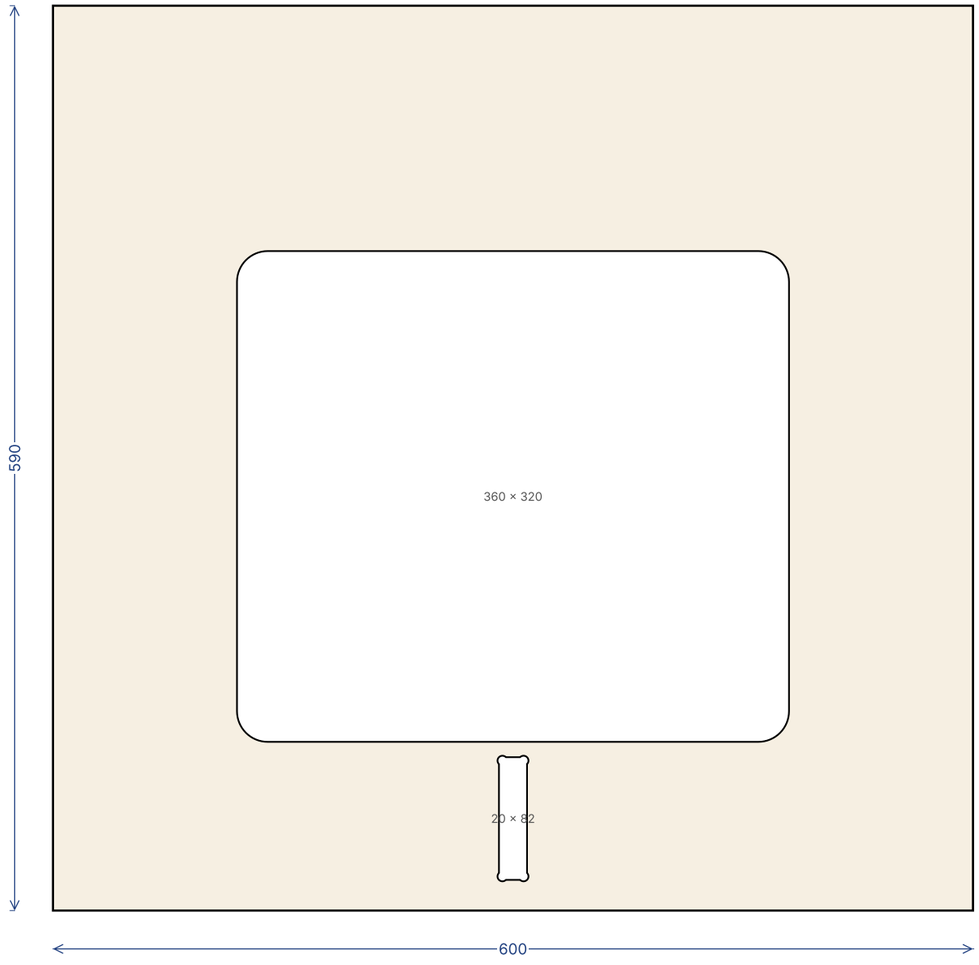
Removable for bolting the backbox down and for service.

#	dia	X	Y	op
1	4.5	9.0	60.0	drill
2	4.5	9.0	279.6	drill
3	4.5	9.0	519.1	drill
4	4.5	591.0	60.0	drill
5	4.5	591.0	279.6	drill
6	4.5	591.0	519.1	drill

P26 Leg frame plate

18 mm plywood | qty 4 | module: legs

scale 1:4 at A3 (origin = lower-left corner)

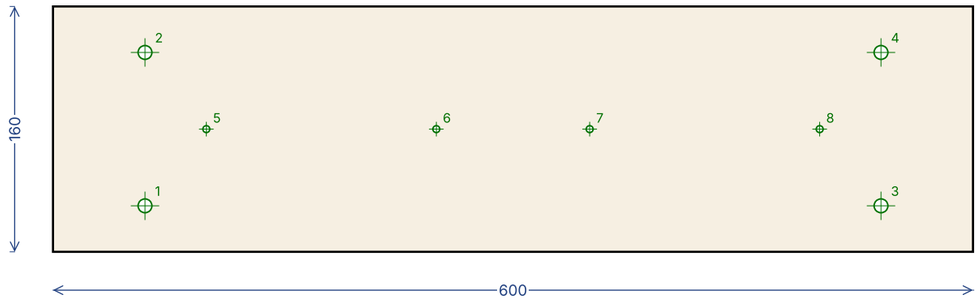


Glue two plates face to face for each leg frame (4 plates make 2 frames).

P27 Leg top plate

18 mm plywood | qty 2 | module: legs

scale 1:4 at A3 (origin = lower-left corner)



Glue and screw down into the frame edge. M8 knob bolts go up through the outer holes.

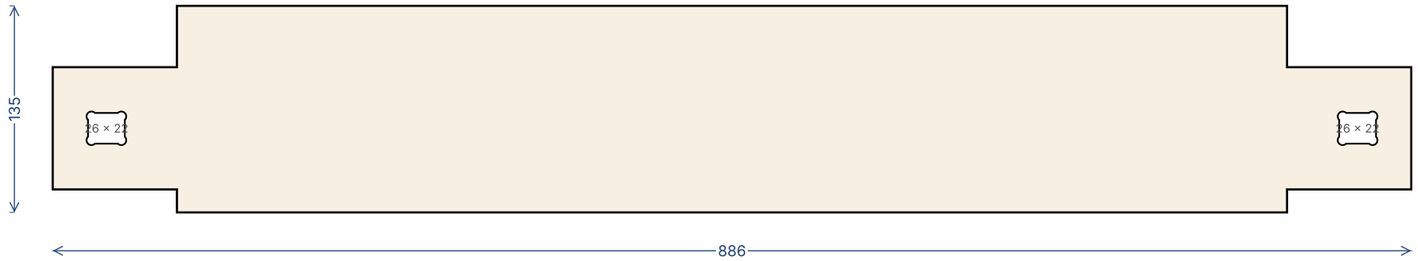
#	dia	X	Y	op
1	9.0	60.0	30.0	drill
2	9.0	60.0	130.0	drill
3	9.0	540.0	30.0	drill
4	9.0	540.0	130.0	drill
5	4.5	100.0	80.0	drill
6	4.5	250.0	80.0	drill
7	4.5	350.0	80.0	drill
8	4.5	500.0	80.0	drill

P28 Leg stretcher

18 mm plywood | qty 1 | module: legs

scale 1:4 at A3 (origin = lower-left corner)

Through-tenons pass the leg frames; a wedge in each end locks it. No tools.

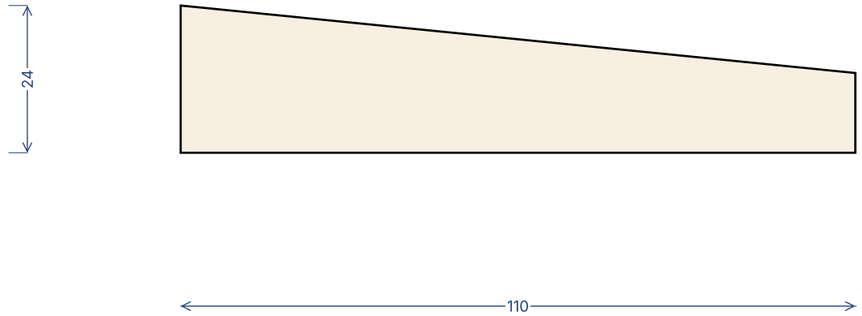


P29 Wedge

18 mm plywood | qty 2 | module: legs | drawn mirrored: shows the cut face (same as the DXF)

scale 1:1 at A3 (origin = lower-left corner)

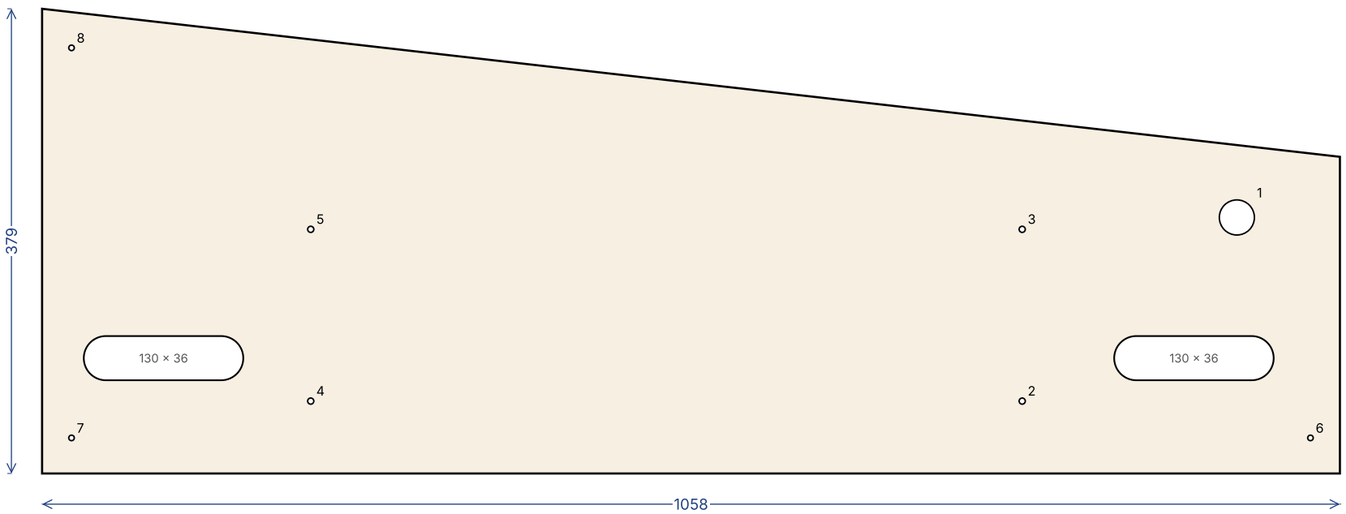
Tap in from either side to pull the leg frame tight.



P30 Skin body side L

3 mm ACM (aluminium composite) + printed vinyl | qty 1 | module: body | drawn mirrored: shows the cut face (same as the DXF)

scale 1:5 at A3 (origin = lower-left corner)



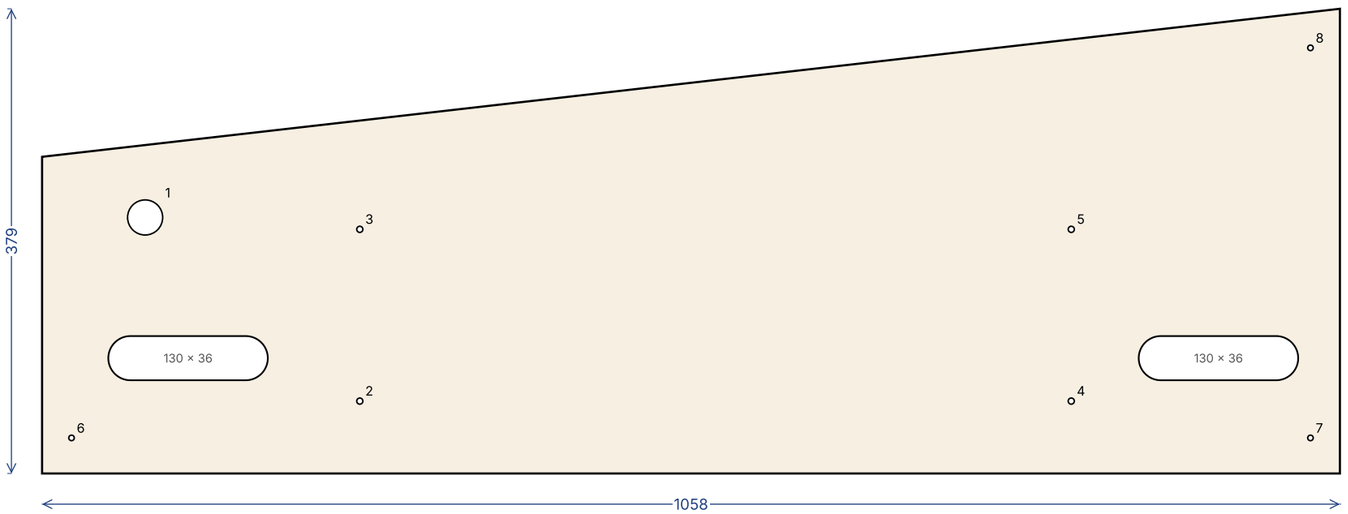
Printed vinyl on 3 mm ACM. Held by the light-panel
standoffs and 3 x M4 screws.

#	dia	X	Y	op
1	28.6	974.0	208.7	profile
2	5.0	799.0	59.0	profile
3	5.0	799.0	199.0	profile
4	5.0	219.0	59.0	profile
5	5.0	219.0	199.0	profile
6	4.5	1034.0	29.0	profile
7	4.5	24.0	29.0	profile
8	4.5	24.0	346.9	profile

P31 Skin body side R

3 mm ACM (aluminium composite) + printed vinyl | qty 1 | module: body

scale 1:5 at A3 (origin = lower-left corner)



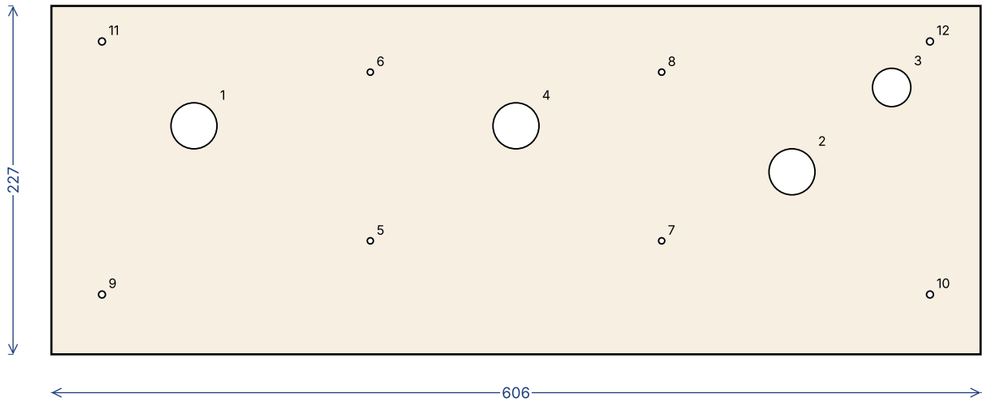
Printed vinyl on 3 mm ACM. Held by the light-panel
standoffs and 3 x M4 screws.

#	dia	X	Y	op
1	28.6	84.0	208.7	profile
2	5.0	259.0	59.0	profile
3	5.0	259.0	199.0	profile
4	5.0	839.0	59.0	profile
5	5.0	839.0	199.0	profile
6	4.5	24.0	29.0	profile
7	4.5	1034.0	29.0	profile
8	4.5	1034.0	346.9	profile

P32 Skin front

3 mm ACM (aluminium composite) + printed vinyl | qty 1 | module: body

scale 1:4 at A3 (origin = lower-left corner)



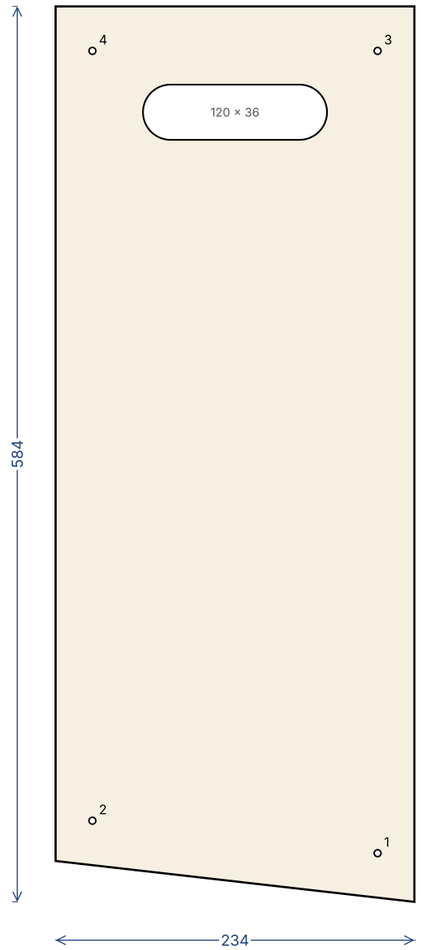
The coin-door plate screws on over it. 4 x M4 at the corners.

#	dia	X	Y	op
1	30.0	93.0	149.0	profile
2	30.0	483.0	119.0	profile
3	25.0	548.0	174.0	profile
4	30.0	303.0	149.0	profile
5	4.0	208.0	74.0	profile
6	4.0	208.0	184.0	profile
7	4.0	398.0	74.0	profile
8	4.0	398.0	184.0	profile
9	4.5	33.0	39.0	profile
10	4.5	573.0	39.0	profile
11	4.5	33.0	204.0	profile
12	4.5	573.0	204.0	profile

P33 Skin backbox side L

3 mm ACM (aluminium composite) + printed vinyl | qty 1 | module: backbox | drawn mirrored: shows the cut face (same as the DXF)

scale 1:4 at A3 (origin = lower-left corner)



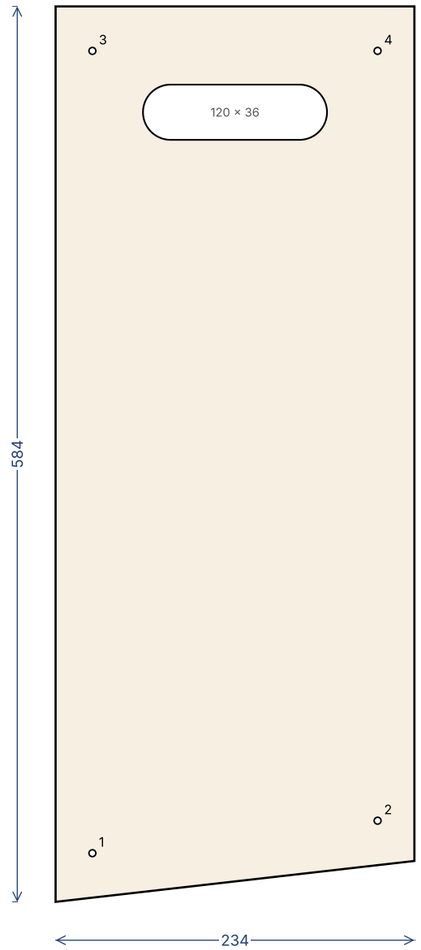
4 x M4 screws into inserts in the backbox side.

#	dia	X	Y	op
1	4.5	210.0	31.7	profile
2	4.5	24.0	52.9	profile
3	4.5	210.0	554.9	profile
4	4.5	24.0	554.9	profile

P34 Skin backbox side R

3 mm ACM (aluminium composite) + printed vinyl | qty 1 | module: backbox

scale 1:4 at A3 (origin = lower-left corner)



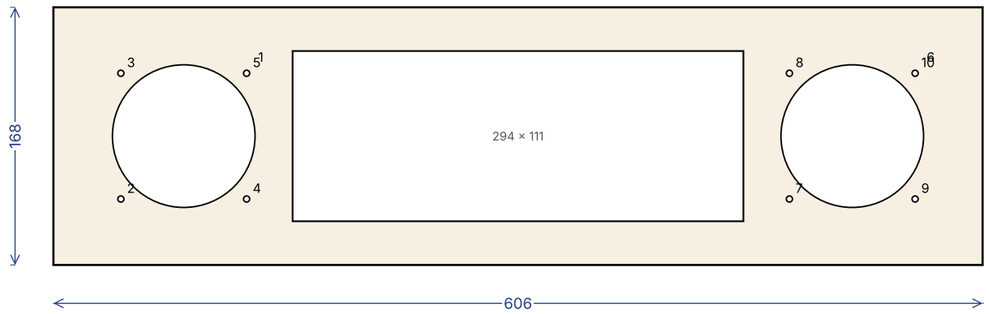
4 x M4 screws into inserts in the backbox side.

#	dia	X	Y	op
1	4.5	24.0	31.7	profile
2	4.5	210.0	52.9	profile
3	4.5	24.0	554.9	profile
4	4.5	210.0	554.9	profile

P35 Skin backbox front

3 mm ACM (aluminium composite) + printed vinyl | qty 1 | module: backbox

scale 1:4 at A3 (origin = lower-left corner)

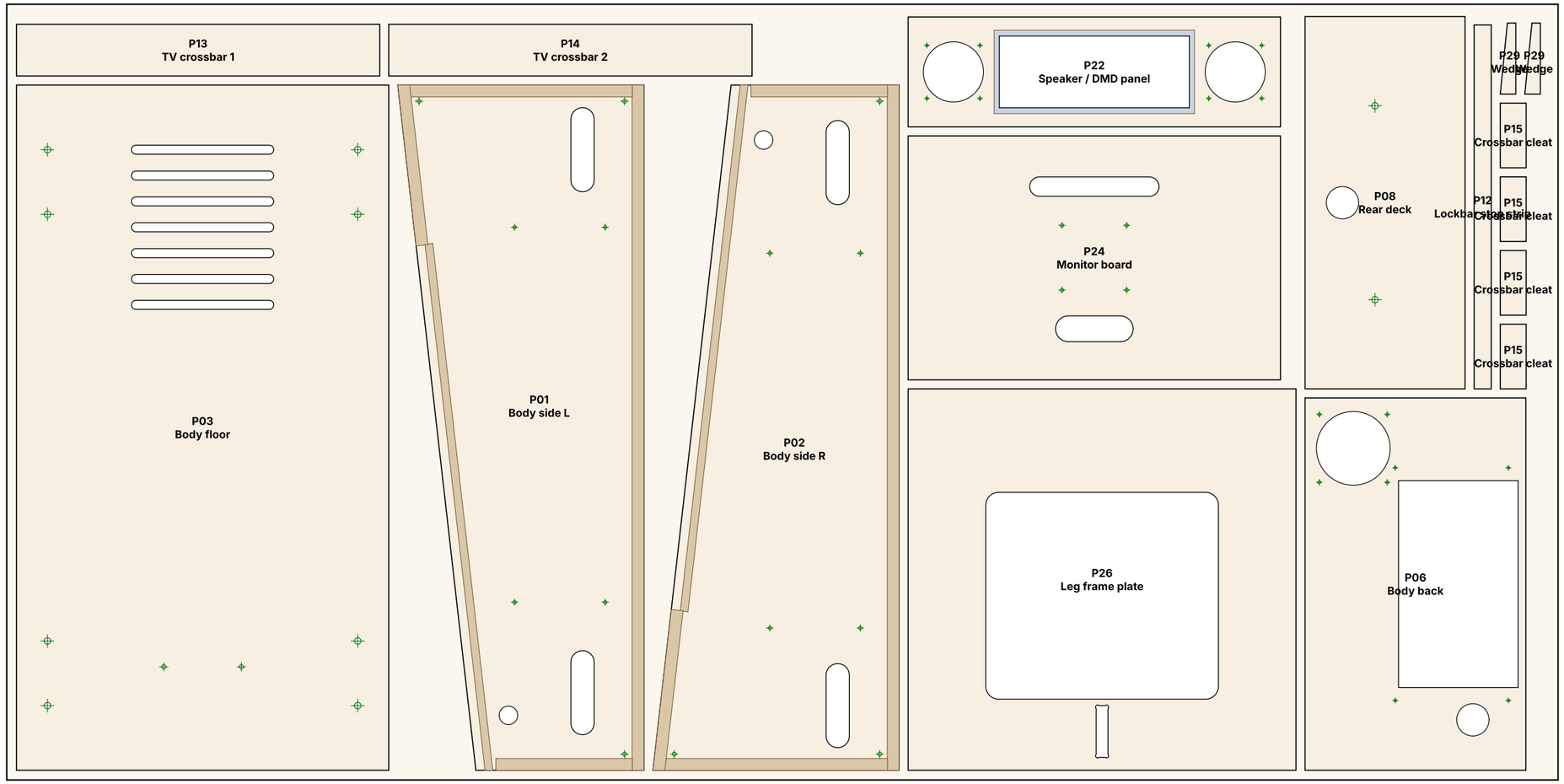


Held by the speaker grille screws.

#	dia	X	Y	op
1	93.0	85.0	84.0	profile
2	4.0	44.0	43.0	profile
3	4.0	44.0	125.0	profile
4	4.0	126.0	43.0	profile
5	4.0	126.0	125.0	profile
6	93.0	521.0	84.0	profile
7	4.0	480.0	43.0	profile
8	4.0	480.0	125.0	profile
9	4.0	562.0	43.0	profile
10	4.0	562.0	125.0	profile

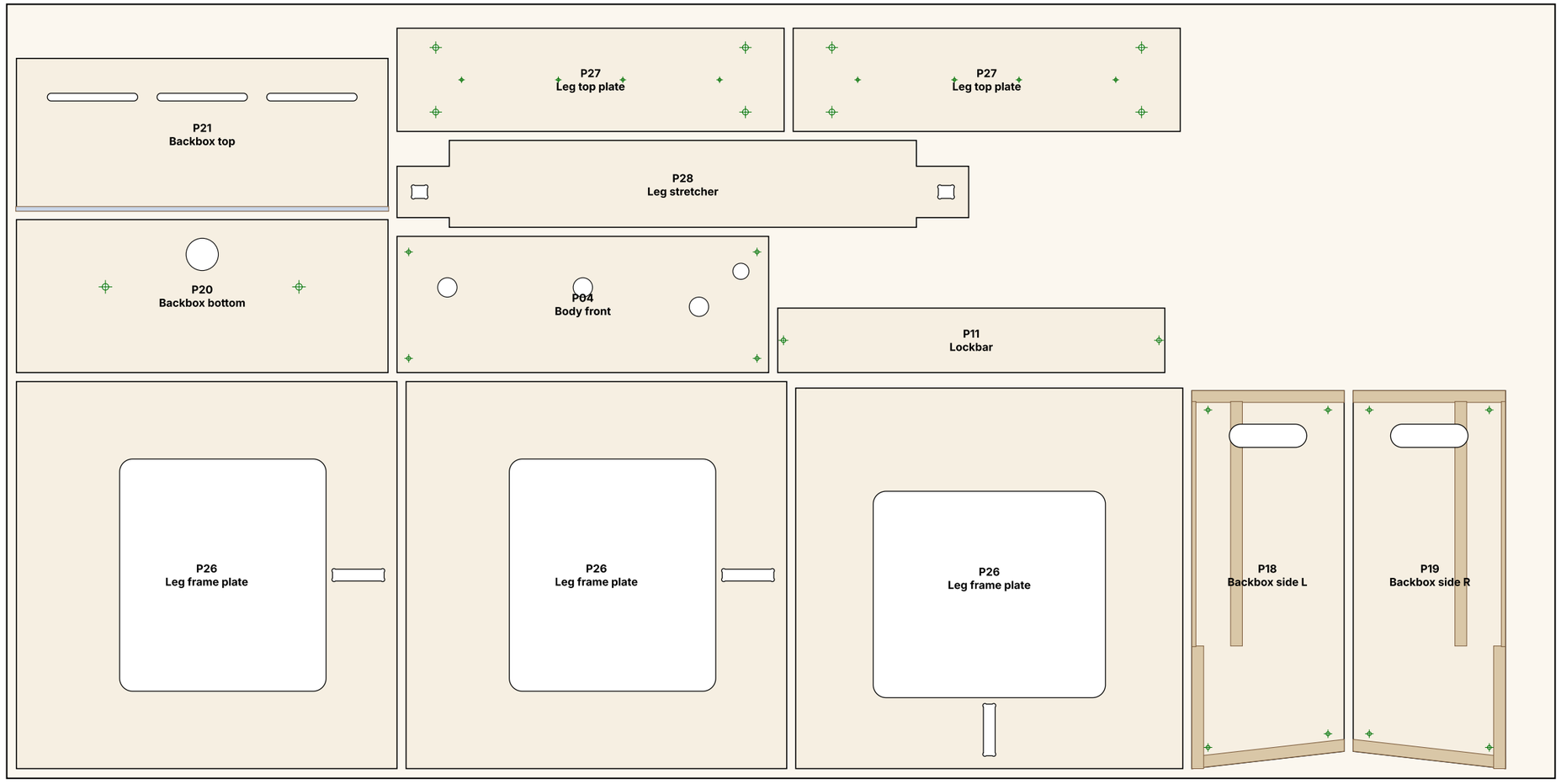
Sheet layout: 18 mm plywood #1 of 2

2400 × 1200 sheet, 14 mm between parts. File: sheets/ply18_1.dxf



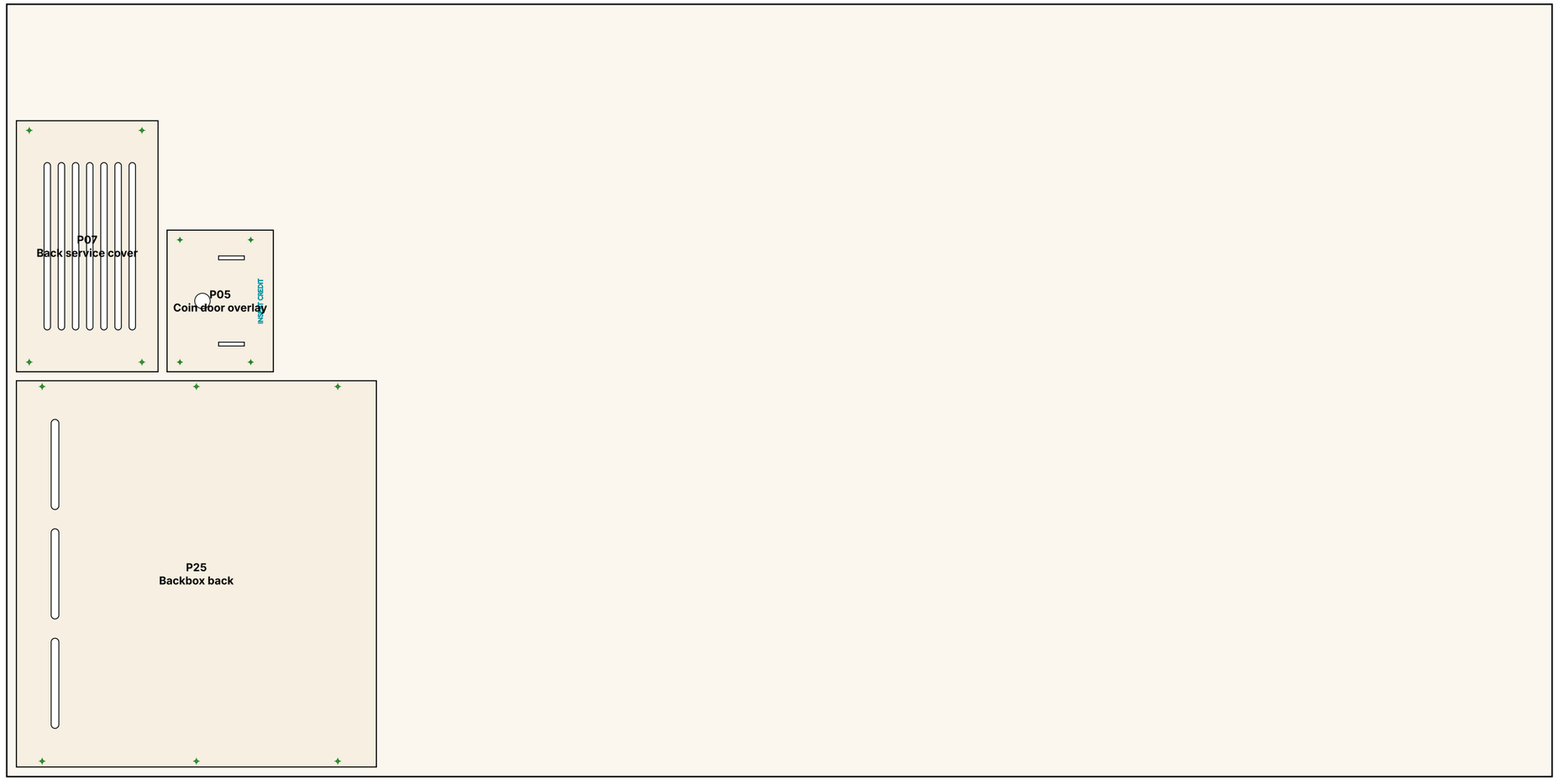
Sheet layout: 18 mm plywood #2 of 2

2400 × 1200 sheet, 14 mm between parts. File: sheets/ply18_2.dxf



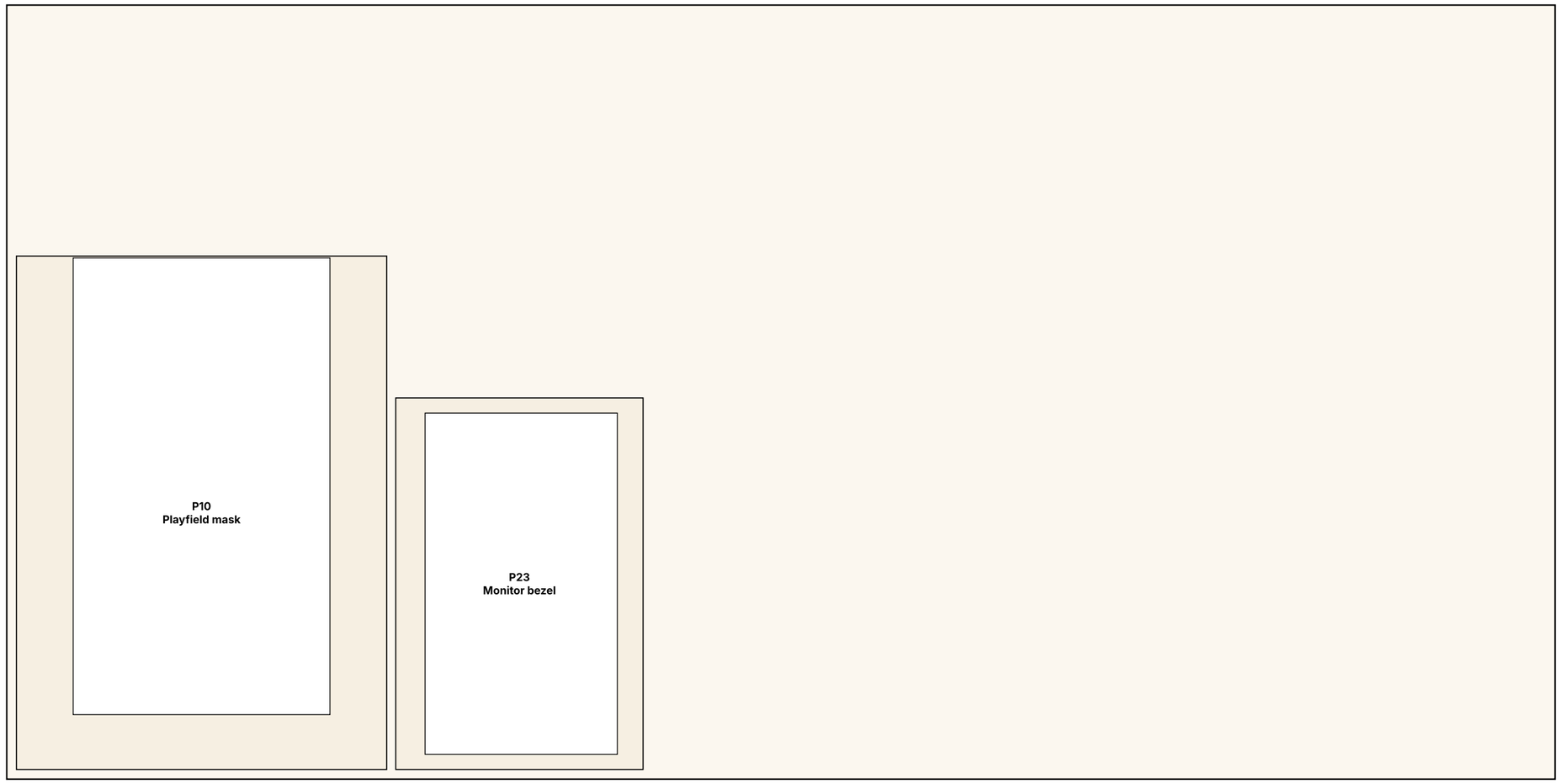
Sheet layout: 6 mm plywood #1 of 1

2400 × 1200 sheet, 14 mm between parts. File: sheets/ply6_1.dxf



Sheet layout: 6 mm MDF #1 of 1

2400 × 1200 sheet, 14 mm between parts. File: sheets/mdf6_1.dxf



Sheet layout: 3 mm clear cast acrylic #1 of 1

1200 × 600 sheet, 14 mm between parts. File: sheets/acrylic3_1.dxf



Sheet layout: 3 mm ACM (aluminium composite) + printed vinyl #1 of 1

2440 × 1220 sheet, 14 mm between parts. File: sheets/acm3_1.dxf

